

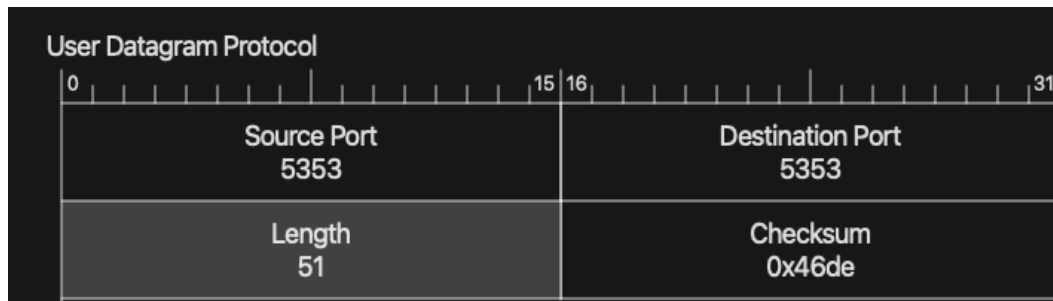
Lab 04 UDP

1. Select one UDP packet from your trace. From this packet, determine how many fields there are in the UDP header. (You shouldn't look in the textbook! Answer these questions directly from what you observe in the packet trace.) Name these Fields.

There are source port, destination port, Length, and checksum in the USP header. So there are 4 fields in the UDP header.

```
✓ User Datagram Protocol, Src Port: 5353, Dst Port: 5353
  Source Port: 5353
  Destination Port: 5353
  Length: 51
  Checksum: 0x46de [unverified]
  [Checksum Status: Unverified]
  [Stream index: 5]
  [Stream Packet Number: 2]
  > [Timestamps]
  UDP payload (43 bytes)
```

2. By consulting the displayed information in Wireshark's packet content field for this packet, determine the length (in bytes) of each of the UDP header fields.
UDP header consists of 4 fields, each of fixed size Source port: 2 bytes, Destination port: 2 bytes, Length: 2 bytes, Checksum: 2 bytes.
UDP Header size = 2 bytes + 2 bytes + 2bytes + 2bytes = 8 bytes



3. The value in the Length field is the length of what? (You can consult the text for this answer). Verify your claim with your captured UDP packet.
The UDP Length is calculated as UDP Header (8 bytes) + Data Payload. From the screenshot above, the total length is 51 bytes, which consists of an 8-byte UDP header and a 43-byte UDP payload.
4. What is the maximum number of bytes that can be included in a UDP payload? (Hint: the answer to this question can be determined by your answer to 2. above)
The UDP length field is 2 bytes(16bits), since it represents the total length of the UDP header + UDP payload, its maximum value is: $2^{16} - 1 - 8(\text{UDP header}) = 65,527$ bytes
Thus, the maximum number of bytes that can be included in a UDP payload is 65527 bytes.

5. What is the largest possible source port number? (Hint: see the hint in 4.)
16-bit field can store values from 0 to $2^{16} - 1$, thus the maximum possible value is $2^{16} - 1 = 65535$, so The largest possible source port number is 65535.
6. What is the protocol number for UDP? Give your answer in both hexadecimal and decimal notation. To answer this question, you'll need to look into the Protocol field of the IP datagram containing this UDP segment (see Figure 4.13 in the text, and the discussion of IP header fields).
The protocol number for UDP is assigned in the Protocol field of the IP header. This field indicates which transport layer protocol is being used. The decimal notation is 17 for the protocol number for UDP and the Hexadecimal notation is: 0X11 which means that when the IP header's Protocol field contains the value 17 (0x11), it signifies that the packet carries a UDP segment.
7. Examine a pair of UDP packets in which your host sends the first UDP packet and the second UDP packet is a reply to this first UDP packet. (Hint: for a second packet to be sent in response to a first packet, the sender of the first packet should be the destination of the second packet). Describe the relationship between the port numbers in the two packets.

UDP sent by my host:

```
▼ User Datagram Protocol, Src Port: 33408, Dst Port: 5353
  Source Port: 33408
  Destination Port: 5353
  Length: 52
  Checksum: 0x357a [unverified]
  [Checksum Status: Unverified]
  [Stream index: 9]
  [Stream Packet Number: 1]
  > [Timestamps]
  UDP payload (44 bytes)
```

UDP replied to my host:

```
▼ User Datagram Protocol, Src Port: 5353, Dst Port: 5353
  Source Port: 5353
  Destination Port: 5353
  Length: 51
  Checksum: 0x46de [unverified]
  [Checksum Status: Unverified]
  [Stream index: 5]
  [Stream Packet Number: 2]
  > [Timestamps]
  UDP payload (43 bytes)
```

The relationship between port numbers is that the source port of the sent message becomes the destination port of the received message, while the destination port of the sent message serves as the source port of the received message.